

Mechanosensitive Modeling of the Mitochondrial Network in HUVECs

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Introduction to Research

Human Umbilical Vein Endothelial Cells (HUVECs) provide a powerful model for investigating mitochondrial responses to mechanical forces in vascular disease. Mechanotransduction, the conversion of mechanical stimuli into biochemical signals, plays a critical role in cardiovascular health and endothelial cell function¹².

Mitochondria, the cellular powerhouses responsible for energy production and calcium homeostasis, exist as dynamic networks that undergo continuous fusion and fission to meet metabolic demands³. These morphological transitions are highly responsive to changes in cellular state³.

Mitochondrial networks can function as dynamic cellular stress sensors, reorganizing in response to various stimuli, including oxidative and metabolic stress³. Although the potential for mitochondrial networks to respond to mechanical forces has been proposed, the mechanisms and functional implications of this response remain poorly understood³.

To address these questions, we developed a model using HUVECs, mitochondrial annotation, and holotomographic microscopy to systematically investigate the mechanosensitive properties of mitochondrial networks.

Background

- Endothelial cells experience constant mechanical forces from blood flow and convert them into biochemical signals via mechanotransduction¹².
- Disturbed or low wall shear stress in blood vessels is linked to endothelial dysfunction and atherosclerosis¹².
- Mitochondria form dynamic networks that undergo fusion and fission to meet cellular energy demands³.
- These networks also reorganize in response to stress³, but their behavior under mechanical stimulation in endothelial cells is poorly understood.
- Quantitative models of mitochondrial network remodeling could reveal new insights into cellular mechanosensing and its role in vascular disease.

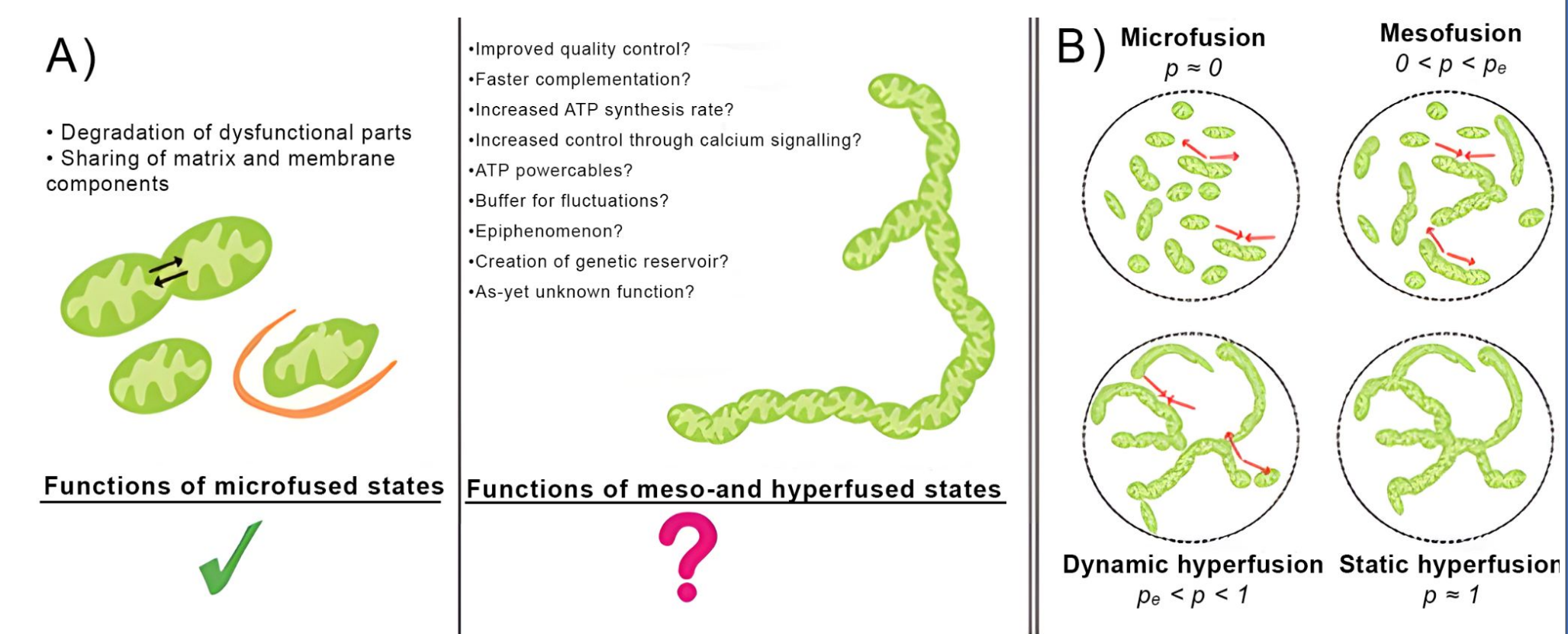


Figure 1. Conceptual diagram of mitochondrial network theory, adapted from Hoitzing et al.³

Methodology

Cell Culture and Flow Stimulation:

HUVECs were cultured at 37°C in a humidified atmosphere containing 5% (v/v) CO₂. Upon reaching confluency, cells were passaged and seeded into either petri dishes for static controls or specialized flow chambers for flow simulation.

Flow chambers were designed to simulate in vivo conditions, exposing HUVECs to controlled wall shear stress representative of vascular flow patterns. Mechanical stimulation protocols were implemented to model the dynamic mechanical environment experienced by endothelial cells in natural conditions.

Holotomographic Microscopy (HTM) and Fluorescent Labeling:

Live-cell imaging was performed using holotomographic microscopy (HTM), which utilizes refractive index measurements to generate three-dimensional cellular images. HTM uses refractive index to differentiate between subcellular organelles by passing light through them and measuring how much the light bends.

Mitochondrial networks were labeled using fluorescent trackers including MitoTracker Deep Red 50μM (for mitochondrial morphology) and MitoSOX 5μM (for reactive oxygen species detection). This dual-modality approach combines label-free HTM imaging with specific fluorescent visualization of mitochondrial dynamics and oxidative stress responses.

Mitochondrial Network Analysis:

Post-acquisition images were processed using specialized image analysis software. Mitochondrial networks were segmented and quantified using both refractive index imaging and fluorescent signal intensity, with analysis focusing on network morphology.

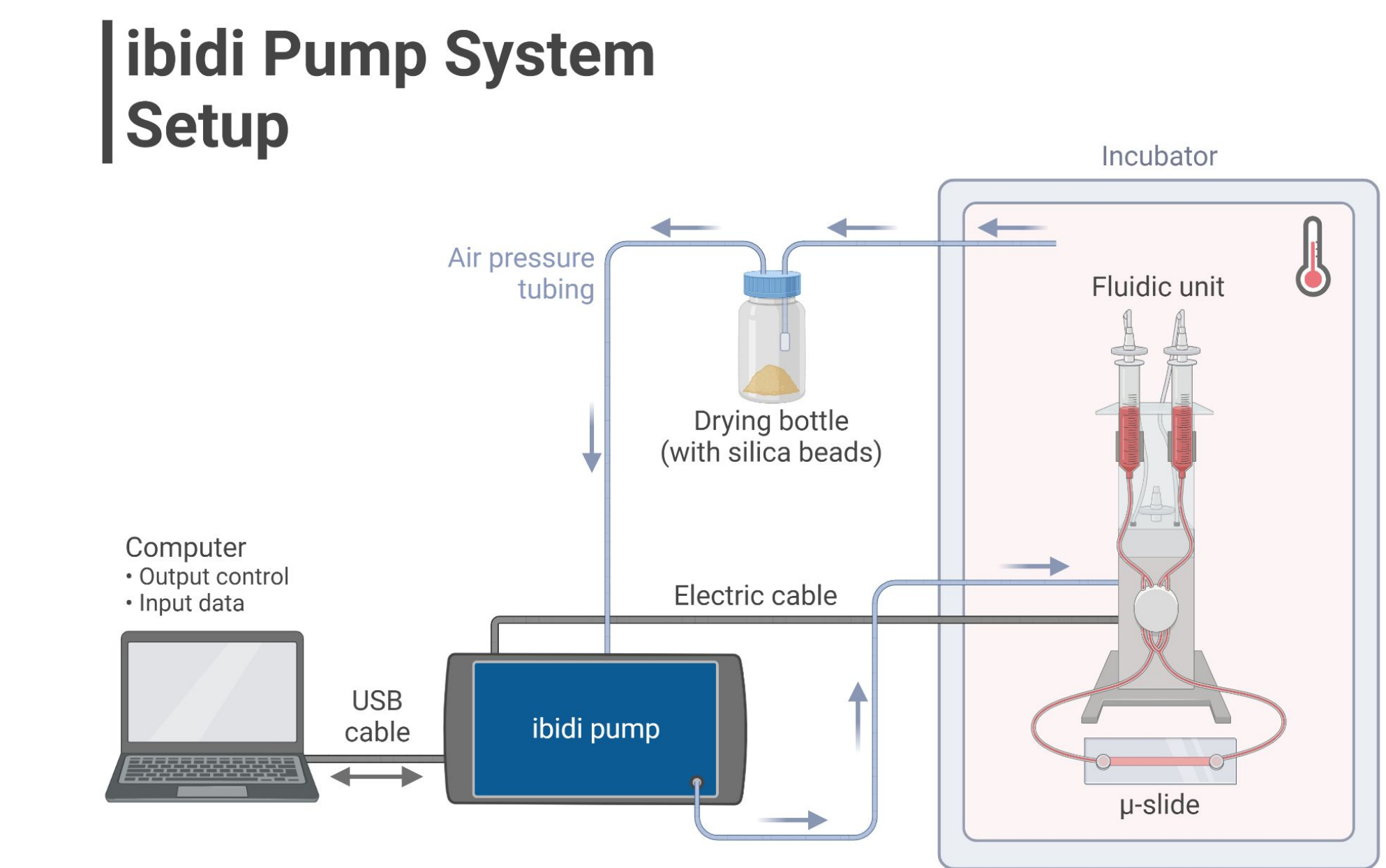


Figure 2. Ibidi pump system schematic. Created with BioRender.com by Mina Nashed⁴.

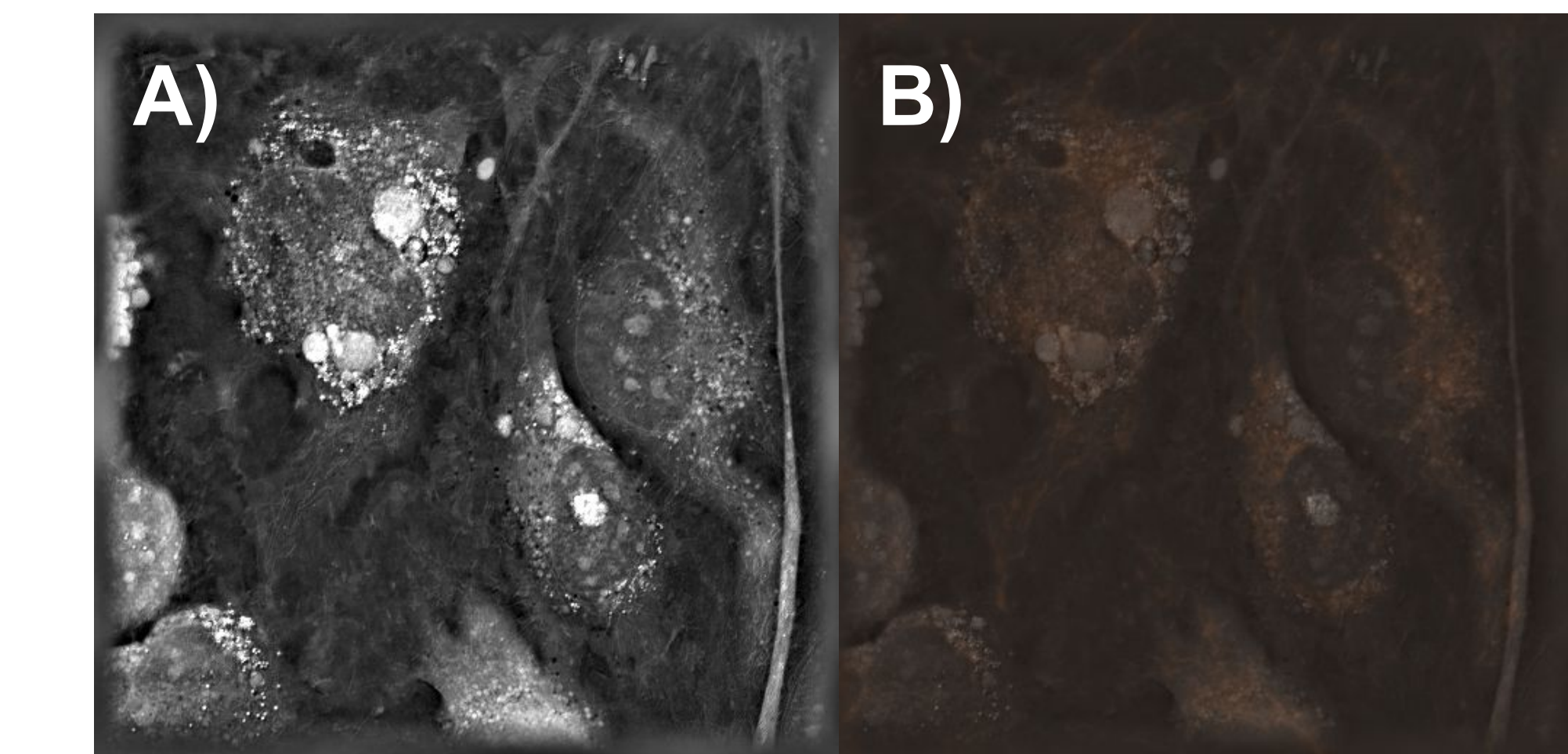


Figure 3. (A) Holotomographic refractive index image of HUVECs. (B) Raw HTM image overlaid with MitoTracker fluorescence in orange.

Results

We successfully developed a comprehensive pipeline for imaging live HUVECs for mitochondrial analysis. Representative results demonstrate our ability to precisely annotate and isolate mitochondrial networks from HTM image sequences.

HTM imaging provided high-contrast visualization of mitochondrial networks within HUVECs (Figure 4A). Manual annotation protocols were established to accurately trace individual mitochondrial structures (red overlay, Figure 4B).

Annotated mitochondrial networks were successfully extracted and converted to grayscale binary images (Figure 4C and 4D), preserving network morphology while removing background cellular structures. This processing pipeline generates clean representations of mitochondrial morphology suitable for analysis by other lab members or computer programs.

Sequential image processing of HTM recordings produces time-resolved datasets of mitochondrial network evolution. These annotated image sequences enable both manual assessment and computational analysis of mitochondrial dynamics and morphological changes in response to mechanical stimulation.

Our annotation methodology provides a robust framework for investigating mechanosensitive mitochondrial responses, establishing the foundation for quantitative analysis of mitochondrial network remodeling under physiological flow conditions.

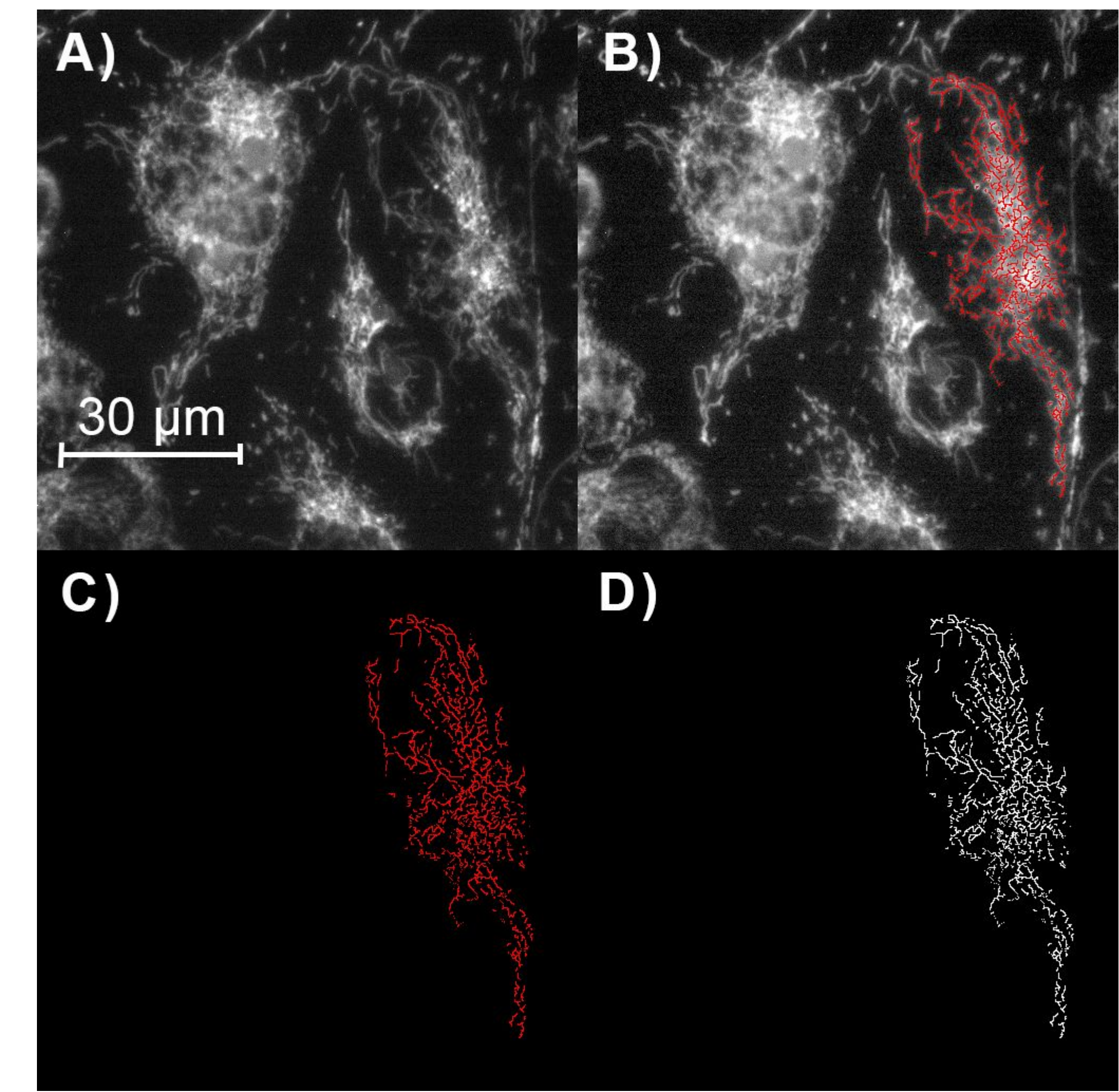


Figure 4: Pipeline for visualizing HUVEC mitochondrial networks, from image acquisition to segmentation.

Future Steps

The images and data we have collected will be used to further develop machine learning algorithms for automated mitochondrial network analysis. This methodology will enable Dr. Jurney's lab to investigate how different mechanical stimuli affect mitochondrial dynamics in real-time, advancing our understanding of mechanotransduction in cardiovascular disease and identifying potential therapeutic targets.



Figure 5. Live Imaging of HUVECs using HTM software.

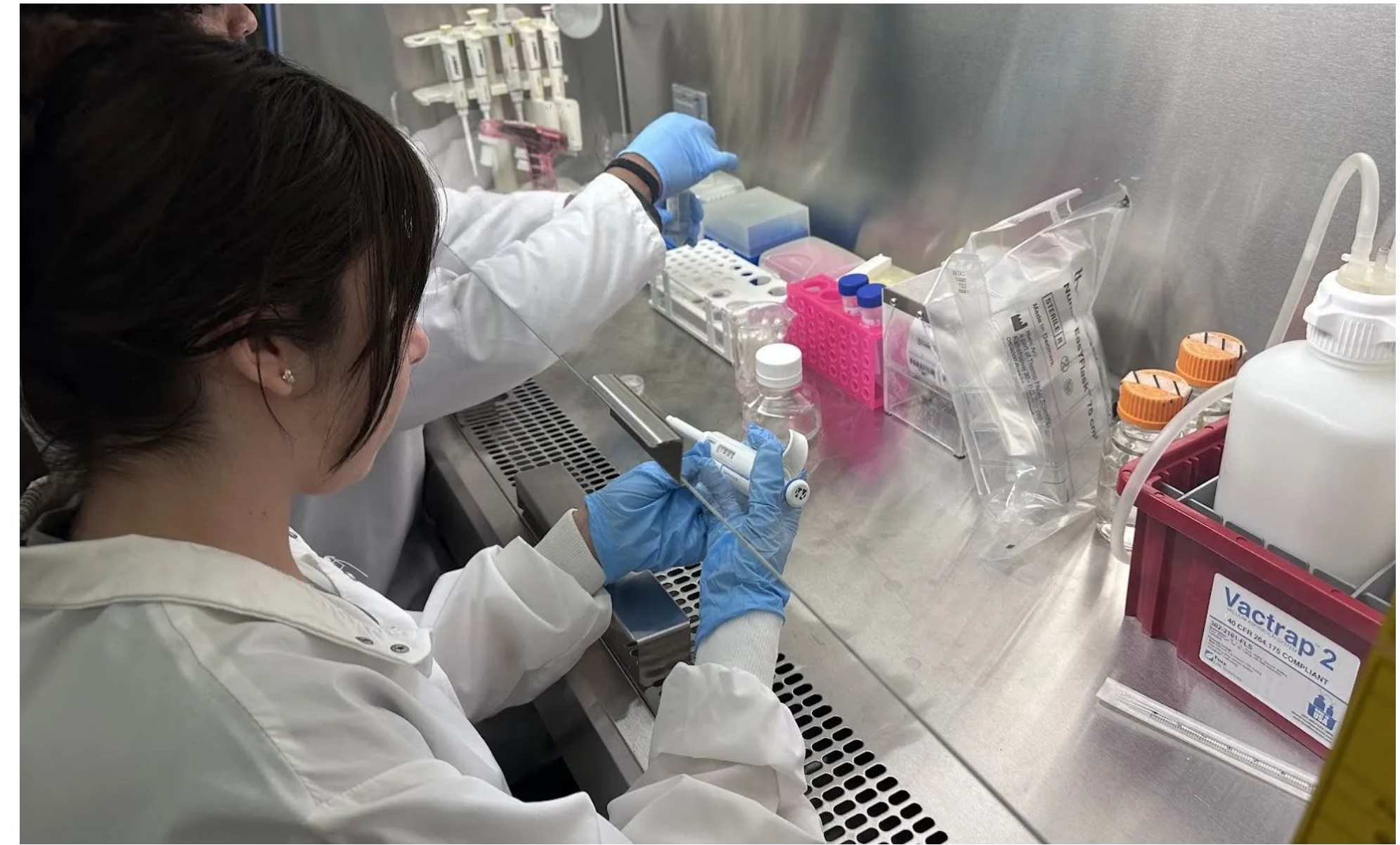


Figure 6. Lab members performing media changes for HUVEC cultures.

References:

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